

Section B

Answer **all** questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

6. (a) Use the information in **Table 1** to answer the following question.

Tick (✓) the boxes next to the **three** correct statements.

[3]

Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto ☐

Mercury is the hottest planet ☐

All the rocky planets have an atmosphere containing carbon dioxide ☐

Mars and the Earth have the same day length ☐

The Earth has the greatest density ☐

The Moon is the smallest planet ☐

(b) Use the information in **Diagram 3** and your knowledge to answer the following questions.

(i) State how the wavelength of the dark line, λ , for the very distant galaxy is different from the same line in the nearby galaxy. [1]

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(ii) State how the frequency of this line for a nearby galaxy compares with the same line in a very distant galaxy. [1]

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(iii) State **one** difference about the motion of the very distant galaxy compared to the nearby galaxy. [1]

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(c) Use the information in **Table 1** to answer the following questions.

(i) Europa is an asteroid. Estimate its temperature and orbital period. [2]

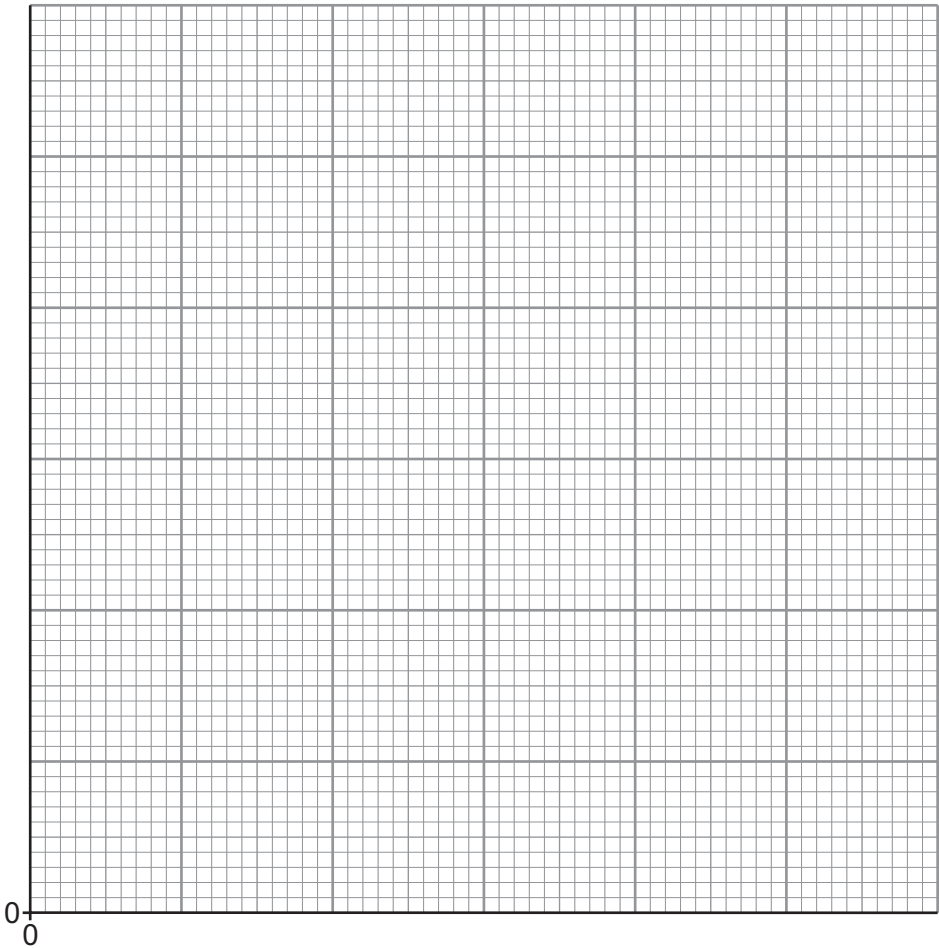
Temperature = °C

Orbital period = days

(ii) Pluto is no longer classed as a planet. State **one** reason why. [1]

(iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line. [4]

Orbital
Velocity
(km/s)



Distance from Sun (AU)

(iv) Orbital velocity is not proportional to the distance from the Sun. Explain how your graph shows this to be true. [2]

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(d) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

(i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

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(ii) State **two** ways in which human activities may be affected by solar flares. [2]

1.

2.

(iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

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(e) Use the information in **Diagram 1** to describe the differences between Aristotle’s model of the Solar System and the 2006 model. [4]

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END OF PAPER